ABERDARE TECHNICAL COLLEGE

INFORMATION COMMUNICATION TECHNOLOGY

MODULE 2,TERM 3

UNIT:COMPUTER REPAIR AND MAINTAINANCE

UNIT CODE: 0619 451 06A

ELEMENT 3:TEST COMPUTER COMPONENTS

TIME 2 hours

Candidate Name	
Admission Number	
Date	
Assessor	

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Requirements

- A computer with a hard disk drive, running Windows
- Internet access to download the software
- A pen and the Results Recording Sheet below
- A flash disk or folder for saving screenshots

Download Link

CrystalDiskInfo official website: https://crystalmark.info/en/software/crystaldiskinfo/

Download page: https://crystalmark.info/en/download/#CrystalDiskInfo

Download only from the official website. Choose either the **installer (.exe)** or the **ZIP (portable)** edition. The ZIP edition needs no installation.

Instructions to the Candidate

1. Attempt all tasks in the order given.
2. Record your results in the Results Recording Sheet. Write exactly what you see on the screen.
3. Take a screenshot at every stage marked "(screenshot)".
4. Do not change any drive settings or delete any files.
5. Work safely: shut down and unplug properly if you handle any hardware, and observe ESD (anti-static) precautions.
6. Inform the assessor when you have finished.

Task 1: Preparation and Safety

1. Put on the required protective gear and prepare your work area.
2. Switch on the computer and log in.
3. Identify the type of storage installed in the computer (HDD or SSD). State how you identified it, for example from Device Manager or Task Manager.
4. Close all unnecessary programs and open the **Disk Management** tool. Note the number of drives and partitions present.
5. State **two** precautions you should observe before testing a storage device.

Task 2: Download and Install the Software

1. Open a web browser and go to the CrystalDiskInfo download link given above.
2. Download the software. Note the name and version of the file you downloaded.
3. Install the software (or extract the ZIP file) and open the program. *(screenshot)*
4. Write down the name of the website you downloaded the software from and why it is important to download from an official source.

Task 3: Identify the Hard Disk Drive

Using the main CrystalDiskInfo window, record the following in the Results Recording Sheet:

1. The drive model name
2. The drive capacity
3. The firmware version
4. The interface type (e.g. SATA)
5. The rotation rate (RPM)
6. The drive letters assigned to it

(screenshot of the main window)

Task 4: Test the Health Status of the Drive

1. Read the **Health Status** shown by the program. Record the status and the percentage, if shown.
2. State what the colour of the health indicator means (Blue/Good, Yellow/Caution, Red/Bad, Grey/Unknown).
3. Record the current **Temperature** of the drive.
4. Record the **Power On Hours** and the **Power On Count**.
5. Explain what these values tell you about how much the drive has been used.

Task 5: Analyse the S.M.A.R.T. Attributes

From the list of S.M.A.R.T. attributes, record the **Current**, **Worst**, **Threshold** and **Raw Values** of the following attributes:

1. Reallocated Sectors Count
2. Current Pending Sector Count
3. Uncorrectable Sector Count
4. Spin Retry Count

5. Temperature

(screenshot of the S.M.A.R.T. attributes list)

Then answer the following:

- (a) What does a rising Reallocated Sectors Count indicate?
- (b) What should you do if the Current value of an attribute falls below its Threshold?

Task 6: Interpret Results and Make Recommendations

1. Based on your results, state whether the drive is **in good condition**, **needs monitoring**, or **should be replaced**. Give two reasons for your decision.
2. State **three** maintenance practices that can extend the life of an HDD.
3. State **two** signs, other than software results, that show an HDD is failing.
4. Explain why you should back up your data before any hardware repair or disk test.

Task 7: Document the Results and Close Down

1. Save the CrystalDiskInfo results by using the **Copy** option in the program or by saving your screenshots.
2. Complete the Results Recording Sheet neatly, with your name and the date.
3. Close the program and leave the work area clean and tidy.
4. Submit the Results Recording Sheet and screenshots to the assessor.

RESULTS RECORDING SHEET

A. Drive Identification

Item	Result Observed
Computer name / number	

Drive type (HDD / SSD)	
Drive model	
Capacity	
Firmware version	
Interface	
Rotation rate (RPM)	
Drive letters	

B. Health Test Results

Item	Result Observed
Health status	
Health percentage	
Temperature (°C)	
Power On Hours	
Power On Count	
Total Host Writes / Reads (if shown)	

C. S.M.A.R.T. Attributes

Attribute	Current	Worst	Threshold	Raw Value
Reallocated Sectors Count				
Current Pending Sector Count				
Uncorrectable Sector Count				
Spin Retry Count				
Temperature				

D. Conclusion

Condition of the drive (tick one): ☐ Good ☐ Needs monitoring ☐ Should be replaced

Reasons: ……

Recommendations: ……

Candidate signature: _____ Date: _____

The trainer is introducing oral questions shortly